

A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

ROSACEA

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

BID Bis In Die (Twice Daily)

GI Gastrointestinal

IPL Intense Pulsed Light

PO Per Os (Oral)

SCTE Stratum Corneum Tryptic Enzyme

SPF Sun Protection Factor

UV Ultraviolet Radiation

Licensed to Shreyansh Salunke (ID: 4477)

OBJECTIVES

- 1 Describe the pathophysiology of rosacea
- 2 Identify signs and symptoms
- 3 Acknowledge the underlying risks and triggers
- 4 Educate patients on non-pharmacological therapies
- 5 Educate patients on pharmacological therapies
- 6 Provide appropriate treatment in pregnancy

OVERVIEW

- Rosacea is a **chronic and progressive cutaneous vascular disorder** – **often misdiagnosed as acne**
- 5th most common skin disorder
- Genetic predisposition found in people of Celtic and North European descent with fair skin
- Approximately 10-50%, or more, of affected patients have **eye involvement** with irritation, dryness, blepharitis, conjunctivitis



PATHOPHYSIOLOGY

- Pathophysiology of inflammation is unclear
- Experts believe it is due to a combination of genetic and environmental factors:

Genetic

- Genetic predisposition towards inflammation found in individuals of Celtic and North European descent with fair skin
- Dysregulation of innate and adaptive immune system

Environmental

- Significant association between *Demodex folliculorum* and rosacea has been reported
- UV exposure may activate molecular pathways

RISK FACTORS

Genetics and family history

- Increased levels of cathelicidin peptides and stratum corneum tryptic enzyme (SCTE)

Increased Parasite exposure

- *Demodex folliculorum* parasite exposure varies by age, skin type, and social determinants of health

Increased UV exposure

Age: 30-50 years

Sex: females (2-3x than men)

Fair skinned individuals of North European descent

COMMON TRIGGERS THAT MAY CAUSE FLARE-UPS



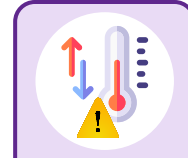
Food and beverages

- Spicy food, vinegar, hot drinks, alcohol



Sun exposure

- UV exposure exacerbates rosacea



Temperature extremes

- Hot, cold, dry, humid, windy climates



Intense exercise

- High intensity workouts trigger skin flushing



Stress

- Emotional stress may worsen rosacea



Medications

- Drugs that cause vasodilation (e.g. calcium channel blockers)



Cosmetic products

- Products with fragrances, alcohol, or acetone

CLINICAL PRESENTATION

Diagnostic Phenotypes	Major Phenotypes	Secondary Features/Phenotypes
• Centrofacial erythema - redness of nose and cheeks (presents as redness on lighter skin tones; dusky/dark brown/grey/purple on darker skin tones) • Phymatous changes - affects the nose, causing tissue to thicken and become bumpy (may also cause widened follicles and a bullous appearance of the nose)	• Flushing - redness in the face and possible feeling of warmth • Papules and pustules - may be mistaken for acne • Telangiectasias - visible blood vessels • Ocular - dryness, burning, stinging, redness, blepharitis (eyelid inflammation), recurrent styes, conjunctivitis, telangiectasia on the eyelids, or discharge affecting the eye	• Burning, stinging • Dryness • Edema (swelling)



Symptoms can vary from mild to severe

DIAGNOSIS

- Diagnosis is based on clinical presentation
 - Based on various phenotypes
- Insufficient evidence may require skin biopsy
- Rule out differential diagnoses:
 - Flushing: carcinoid syndrome, perimenopause, medullary carcinoma of the thyroid
 - Centrofacial erythema: photodamage, systemic lupus erythematosus, seborrheic dermatitis, psoriasis
 - Papules/pustules: acne vulgaris (characterized by presence of comedones) and folliculitis
 - Phymatous changes: nonmelanoma skin cancer

GOALS OF THERAPY

- ✓ Identify and avoid triggers
- ✓ Manage signs and symptoms
- ✓ Reduce number and severity of recurrences
- ✓ Improve quality of life

NON-PHARMACOLOGICAL MEASURES

Avoid Triggers

- Triggers may include UV light, hot foods and beverages, spices, alcohol

Laser treatment

- Can significantly improve telangiectasias

Skin care

- Gentle, fragrance-free, non-soap, non-astringent products

Skin protection

- Daily use of broad-spectrum SPF (30+) and protective clothing

Cosmetic camouflage

- Green-tint foundation overlaid with flesh-coloured foundation

PHARMACOLOGICAL MEASURES

FACIAL REDNESS, FLUSHING, SKIN DRYNESS/SENSITIVITY

Start with topical agents

Topical therapy:
brimonidine,
metronidazole or
azelaic acid

Inadequate response in 8-12 weeks?

Intense pulsed light
(IPL) or vascular laser*

Inadequate response?

PO doxycycline

*IPL or vascular laser is a first-line option if erythema is due to telangiectasias
Consider dermatologist referral if moderate/severe

PHARMACOLOGICAL MEASURES

PHYMATOUS ROSACEA

Mild to moderate cases

Topical retinoid or PO antibiotics* (doxycycline or tetracycline)

* PO antibiotics may be useful particularly if there is an inflammatory component

Inadequate response in 8-12 weeks?

Topical retinoid + PO antibiotics

Inadequate response?

Isotretinoin

Severe cases: surgical/electrosurgical/laser ablation; isotretinoin if patient refuses procedural therapy.

**Weak evidence for the treatment options in phymatous rosacea*

ROSACEA MANAGEMENT

PAPULOPUSTULAR ROSACEA

Start with topical agents

Topical azelaic acid,
metronidazole, or
ivermectin

If moderate/severe*, trial for 8-12 weeks

Add PO antibiotics
(doxycycline or
tetracycline) to topical
therapy

Inadequate response?

If moderate/severe:
Low-dose PO
isotretinoin

*Consider dermatologist referral if moderate/severe

PHARMACOLOGICAL MEASURES

OCULAR ROSACEA

For mild cases, start with hygiene+ tears

Eyelid hygiene
(wash lid/lashes with
warm water or baby
shampoo)
AND artificial tears

Persistent or worsening?

Add **PO antibiotics**
(**doxycycline** or
tetracycline)

Inadequate response?

Refer to **ocular specialist***
and/or cyclosporine
drops

Moderate/severe cases: refer to eye specialist*, and treat with eyelid hygiene, artificial tears, AND PO antibiotics



*Ophthalmologist preferred, if not available refer to optometrist

SPECIFIC THERAPIES

Rosacea	Topical	Systemic	Others
Redness, Flushing, Skin dryness/ Sensitivity	Brimonidine, metronidazole, azelaic acid		- IPL or vascular Laser - Cosmetic camouflage
Papulopustular	- Metronidazole - Azelaic acid - Ivermectin	- Doxycycline, tetracycline, minocycline* - Isotretinoin	
Phymatous	Topical retinoid	- Doxycycline, tetracycline, minocycline* - Isotretinoin	Surgical/ electrosurgical/ laser ablation
Ocular	- Artificial tears - Cyclosporine drops - Ophthalmic antibiotic ointment	Doxycycline, tetracycline, minocycline*	Eyelid hygiene: cleanse lids/lashes BID with baby shampoo on wet washcloth

*Minocycline: Weak evidence and serious side effects, consider if doxycycline/tetracycline fail

DRUG TABLES

Dosing is FYI

Improvement timeline varies; topical therapies may be continued indefinitely (due to relapse on discontinuation)

Class and dosing	Comments	Side Effects
Dicarboxylic acids: Azelaic acid (Finacea®) 15% gel applied BID	1st line therapy in mild-moderate rosacea with papules/pustules - Used as combination therapy in severe cases Onset 4-8 weeks - Can be used for long-term therapy - Antibacterial, anti-inflammatory, and anti-oxidant properties	Skin irritation (moisturizing can help improve), burning, stinging, pruritus, erythema, contact dermatitis
Nitroimidazoles: Metronidazole (MetroGel®) 0.75% gel BID or 1% cream daily, applied sparingly	1st line therapy in mild-moderate rosacea with papules/pustules Least irritating in practice – drug of choice - Used as combination therapy in severe cases Onset 2-4 weeks, can take 8-9 weeks for full benefit	Local burning, stinging, pruritus, skin irritation
Alpha ₂ -adrenergic agonists: Brimonidine tartrate (Onreltea®) 0.33% gel applied daily to the 5 areas of the face (forehead, chin, nose, cheeks)	- Effective only on general facial redness, works by constricting blood vessels - Used for erythema or flushing, not for lesions Onset 30 min; peak in 3 h, effects lasts up to 12 h	Erythema, flushing, headache, burning, contact dermatitis Rebound erythema may occur on withdrawal. May result in more pronounced (not worsened) telangiectasia due to less surrounding redness
Anti-parasitics: Ivermectin 1% cream (Rosiver®) applied daily to the 5 areas of the face (forehead, chin, nose, each cheek)	- Choice for inflammatory lesions - Used in papulopustular rosacea - Can be used long-term up to 1 year	Localized burning, skin irritation, dry skin

SYSTEMIC THERAPY

Dosing is FYI

Oral antibiotics are used to gain control of symptoms (anti-inflammatory properties) in combination with topical agents

Class and dosing	Comments	Side Effects
Tetracyclines: Doxycycline MR (Apprilon®) 40 mg PO daily for 12 weeks Doxycycline 100 mg PO daily for 12 weeks Minocycline 50-100 mg PO daily for 6-8 weeks Tetracycline 500 mg PO BID for 2 weeks, then 500 mg PO daily until controlled, taper to 250 mg daily for 3-4 weeks	• First-line therapy in moderate to severe rosacea in combination with topical agents or topical treatment failure • Used in phymatous and ocular rosacea • Used for 3 months then re-evaluated • Contraindicated in pregnancy, patients on concurrent oral isotretinoin, children <8 years (doxycycline, <12 years (tetracycline), <13 years (minocycline) • Avoid divalent cations (e.g. Mg²⁺, Ca²⁺, Zn²⁺) 2 h before or 4 h after • Relapse may occur if treatment is discontinued	Increased resistance, photosensitivity, GI effects, yeast overgrowth Associated with pseudotumor cerebri (benign intracranial hemorrhage) and azotemia Minocycline: lupus-like syndrome (rare), hyperpigmentation, autoimmune hepatitis
Retinoids: Isotretinoin (Accutane®, Clarus®, Epuris®) 0.2 – 0.3 mg/kg/day PO for 5-6 months	• Used for treatment-resistant rosacea • Low-dose and intermittent-dose regimens may reduce the frequency and severity of adverse events • Teratogenic; contraindicated in pregnancy. Treatment duration in rosacea is longer than for acne, caution in females of childbearing potential • Females must use 2 forms of contraception • Microdosed progesterone (minipill) not compatible • Avoid pregnancy 1 month before, during treatment, and 1 month after discontinuing • Paternal exposure at conception is not expected to pose a risk to a pregnancy	• Photosensitivity, skin dryness, cheilitis, pruritus, erythema, contact dermatitis, conjunctivitis, hepatitis, increased risk of depression/suicide • Rare cases of pseudotumor cerebri

SPECIAL POPULATIONS- PREGNANCY AND BREASTFEEDING

Pre-Pregnancy

Oral isotretinoin is teratogenic and **must be stopped at least 1 month prior to conception**

Pregnancy

Topicals favoured due to minimal systemic absorption (**azelaic acid, metronidazole**)

Avoid tetracyclines and isotretinoin

Breastfeeding

Topical options preferred (e.g. **metronidazole** and **azelaic acid**)

Avoid isotretinoin and tetracyclines while breastfeeding

MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Facial erythema	Improvement	Few hours with brimonidine Other agents can take weeks
Papules, pustules	Early improvement —————→ Adequate response —————→	4-8 weeks 8-12 weeks

SAFETY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Skin irritation and dryness	Prevent or minimize	Duration of therapy
Photosensitivity		

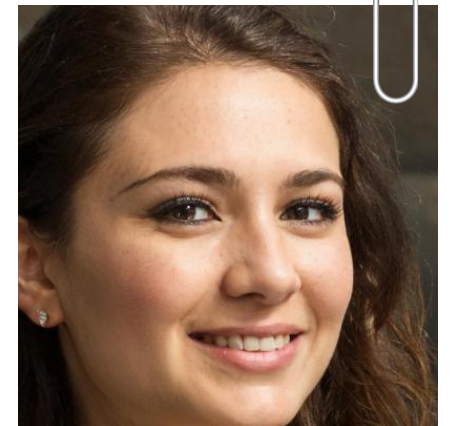
TAKEAWAY

- Patient education on determining triggers is important
- **Avoid long-term use of topical corticosteroids**
 - Can precipitate or worsen rosacea
- Ensure sun protection (sunscreen with SPF $\geq$ 30)
- Assess for ocular involvement and progression
- Consider episodic therapy if mild, remitting disease
- Improvement timeline generally varies, but topical therapies may need to be continued indefinitely (due to relapse on discontinuation)
- Oral antibiotics are used to gain control of symptoms (anti-inflammatory properties) in combination with topical agents
- Consider shifting to topical maintenance therapy once control has been achieved
- Inform patients that they may require life-long medication therapy, as treatments are not curative

CASE

MK is a 35-year-old female presenting to her family doctor with facial flushing and redness around her nose, chin, and cheeks. She is not experiencing any other symptoms. She has been going through a lot in her personal and professional life and has been very stressed, so she wonders if her symptoms can be attributed to this. She feels like the changes in her appearance are also adding to her stress, and she hopes to resolve this issue. Her past medical history is insignificant. She takes Vitamin D 1000 units daily and Alesse® 28 (levonorgestrel 100 mcg and ethinyl estradiol 20 mcg) one tablet PO daily (last taken one month ago). She drinks a glass of wine daily, enjoys eating spicy food, and spends time outdoors. She has no known allergies.

1. Which of the following phenotypes is predominant in MK's rosacea?
 - a) Phymatous changes
 - b) Ocular manifestation
 - c) Papules and pustules
 - d) Centrofacial erythema



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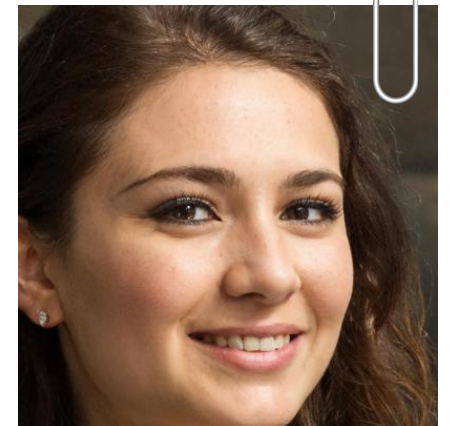
2. Which of the following is NOT a non-pharmacological measure to recommend to MK?
- a) Avoid triggers such as spicy food and alcohol
 - b) Wear sunscreen and protective clothing when going outside
 - c) Avoid non-soap cleansers and fragrance-free moisturizers
 - d) Use green-tinted foundation



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3. Which of the following is the MOST appropriate initial therapeutic option for MK?
- a) Topical brimonidine
 - b) Topical azelaic acid
 - c) Topical metronidazole
 - d) Topical retinoid



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4. MK reveals to the doctor that she is pregnant. Which of the following statements is most correct regarding her therapeutic options?
- a) Topical metronidazole is not considered safe during the first trimester
 - b) Tetracyclines may be considered during the 2nd and 3rd trimester
 - c) Isotretinoin can be used up to the 1st trimester of pregnancy
 - d) Topical azelaic acid is absorbed minimally and is an option in pregnancy



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CHANGE LOG

March 2025

- Formatting changes throughout

August 2025

- Formatting and grammar changes throughout
- Slide 3: Added legend
- Slide 5: Updated statistic
- Slide 7: Changed 'gender' to 'sex'
- Slide 10: Updated differential diagnoses
- Slides 13-17: Updated algorithms and summary table to align with Canadian Dermatology Association 2016 guidelines on Rosacea
- Slide 18: Added peak effect time of brimonidine, added 'dosing is FYI'
- Slide 19: Updated age contraindication for tetracycline antibiotics, added PO to doses
- Slide 26: Updated references

September 2025

- Formatting changes throughout
- Slide 9: Removed 'FYI', added to definitions (erythema, phyma, ocular)
- Slide 11: Added 'improve quality of life' to Goals of Therapy
- Slides 15, 16: Updated algorithms
- Slide 18: Updated drug table (metronidazole dosing, brimonidine side effects)
- Slide 19: Updated drug table (tetracyclines, retinoids)
- Slide 21: Added monitoring parameters
- Slide 27: Updated references

October 2025

- Slide 25: Changed question to 'MOST' appropriate therapy

February 2026

- Slide 2: Updated NAPRA competencies
- Slide 14: Minor edits to treatment algorithm of phymatous rosacea
- Slide 22: Edited the last takeaway point